

MultiLabelSoftMarginLoss#

<https://pytorch.org/docs/stable/generated/torch.nn.MultiLabelSoftMarginLoss>.

Description:

Creates a criterion that optimizes a multi-label one-versus-all loss based on max-entropy, between input `xxx` and target `yyy` of size (N, C) . For each sample in the minibatch: where $i \in \{0, \dots, x.\text{nElement}() - 1\}$, $y[i] \in \{0, 1\}$. `weight` (Tensor, optional) – a manual rescaling weight given to each class. If given, it has to be a Tensor of size `C`. Otherwise, it is treated as if having all ones. `size_average` (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field `size_average` is set to `False`, the losses are instead summed for each minibatch. Ignored when `reduce` is `False`. Default: `True` `reduce` (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on `size_average`. When `reduce` is `False`, returns a loss per batch element instead and ignores `size_average`. Default: `True`

Function Signature

```
class torch.nn.MultiLabelSoftMarginLoss(weight=None,
size_average=None, reduce=None, reduction='mean')[source]#
```

Parameters

Shape:

```
forward(input, target)[source]#
```

Return type

Returns:

Tensor

Detailed Documentation

Documentation

Creates a criterion that optimizes a multi-label one-versus-all loss based on max-entropy, between input `xxx` and target `yyy` of size (N,C) (N, C) (N,C) . For each sample in the minibatch:

where $i \in \{0, \dots, x.nElement()-1\} \in \left\{0, \dots, \text{x.nElement}() - 1\right\}$ $y[i] \in \{0, 1\}$ $y[i] \in \left\{0, 1\right\}$.

`weight` (Tensor, optional) – a manual rescaling weight given to each class. If given, it

has to be a Tensor of size C. Otherwise, it is treated as if having all ones.

`size_average` (bool, optional) – Deprecated (see `reduction`). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field `size_average` is set to `False`, the losses are instead summed for each minibatch. Ignored when `reduce` is `False`. Default: `True`

`reduce` (bool, optional) – Deprecated (see `reduction`). By default, the losses are averaged or summed over observations for each minibatch depending on `size_average`. When `reduce` is `False`, returns a loss per batch element instead and ignores `size_average`. Default: `True`

`reduction` (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: `size_average` and `reduce` are in the process of being deprecated, and in the meantime, specifying either of those two args will override `reduction`. Default: 'mean'

Input: (N,C)(N, C)(N,C) where N is the batch size and C is the number of classes.

Target: (N,C)(N, C)(N,C), label targets must have the same shape as the input.

Output: scalar. If reduction is 'none', then (N)(N)(N).

Runs the forward pass.